IGCSE 0607 Practice Questions — E7.3 — Combined Questions & Markscheme

Exact trigonometric values

Total Questions: 15 — **Total Marks:** 100 Marks

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- **E7.3** Exact trigonometric values
  → **Paper 2 (Non-Calculator)** *(15 Qs)*

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1 Without using a calculator, answer the following questions.

(a) Write down the exact value of $\cos 30°$.

..... **[1]**

(b) Show that

$$\frac{1-\sin 30°}{\tan^2 60°} = \frac{1}{6}$$

[2]

2 Without using a calculator, write down the exact value of each of the following expressions.
You must show all your working.

(a) $\sin 90^\circ + \cos 0^\circ$

..... **[1]**

(b) $\tan 45^\circ \times \sin 30^\circ$

..... **[1]**

(c) $\dfrac{\tan 60^\circ}{\cos 30^\circ}$

..... **[2]**

3 **(a)** Work out the exact value of

$2\sin 60° \times \tan 30°$

..... **[2]**

(b) Find the value of x, where $0° \leq x \leq 90°$, such that

$\sqrt{2}\cos x = \tan 45°$

$x =$ ..... ° **[2]**

4 Without using a calculator, work out the exact value of each of the following.

(a)

$\sin 30° + \cos 60°$

..... **[1]**

(b)

$\tan 60° \times \sin 60° - \cos 0°$

..... **[3]**

5 Do not use a calculator in this question.

(a) Work out the exact value of

(i) $\sin 30° + \cos 60°$,

..... **[2]**

(ii) $\dfrac{\tan 60°}{\cos 30°} - (\sin 45°)^2$.

..... **[3]**

(b) For $0° \leq x \leq 90°$, solve the equation

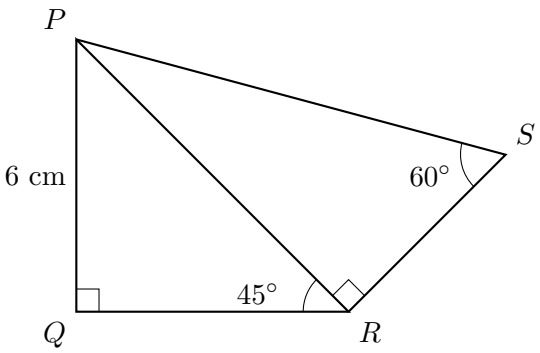
$$2\cos x - \sqrt{3} = 0$$

$x =$ ..... ° **[2]**

(c) Given that $\tan y = \cos 0° - \sin 90°$ and $0° \leq y \leq 90°$, find the value of y.

$y =$ ..... ° **[2]**

6 The diagram shows two right-angled triangles, PQR and PRS.
$PQ = 6$ cm, angle $PQR = 90°$ and angle $PRQ = 45°$.
Angle $PRS = 90°$ and angle $PSR = 60°$.



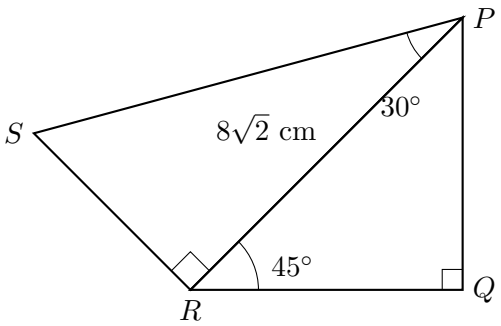
(a) Calculate the exact length of PR.

..... cm **[2]**

(b) Calculate the exact length of PS.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

..... cm **[3]**

7 The diagram shows two right-angled triangles, PQR and PRS.
Angle $PQR = 90°$ and angle $PRS = 90°$.
Angle $PRQ = 45°$ and angle $SPR = 30°$.
$PR = 8\sqrt{2}$ cm.



NOT TO SCALE

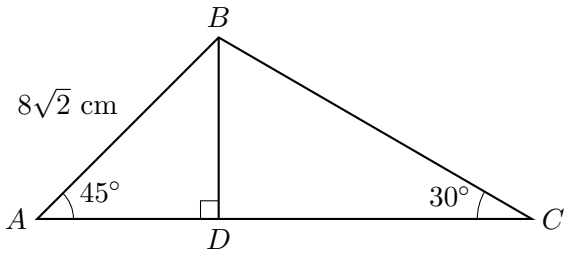
(a) Without using a calculator, show that the exact length of PQ is 8 cm.

[2]

(b) Without using a calculator, work out the exact length of PS.
Give your answer in the form $\frac{a\sqrt{b}}{c}$, where a, b and c are integers.

$PS =$ ..... cm **[3]**

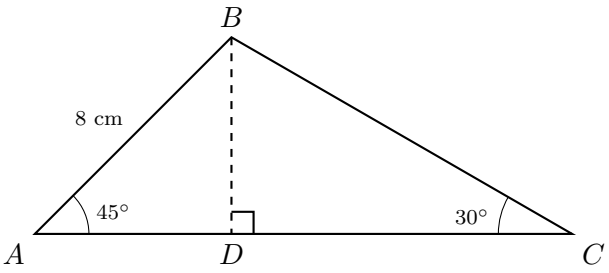
8 The diagram shows a triangle ABC with a point D on the side AC.
BD is perpendicular to AC.
$AB = 8\sqrt{2}$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.



Calculate the exact perimeter of triangle ABC.
Give your answer in the form $a + b\sqrt{2} + c\sqrt{3}$ where a, b, and c are integers.

Perimeter = ... cm **[6]**

9 The diagram shows a triangle ABC with $AB = 8$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.
BD is perpendicular to AC.



(a) Find the exact length of BD.
Give your answer in its simplest surd form.

$BD =$ ..... cm **[2]**

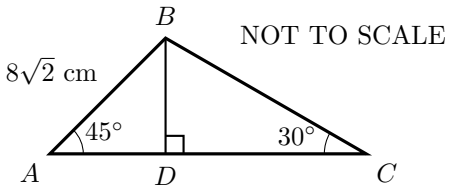
(b) Find the exact length of BC.
Give your answer in its simplest surd form.

$BC =$ ..... cm **[2]**

(c) Show that the exact length of AC is $(4\sqrt{2} + 4\sqrt{6})$ cm.

[2]

10 The diagram shows triangle ABC with BD perpendicular to AC, where D lies on AC.
$AB = 8\sqrt{2}$ cm, angle $BAD = 45°$, and angle $BCD = 30°$.



(a) Show that $BD = 8$ cm and $AD = 8$ cm.

[2]

(b) Calculate the exact length of CD, giving your answer in the form $k\sqrt{3}$, where k is an integer.

$CD =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Area = ... cm^2 **[3]**

(d) Calculate the exact perimeter of triangle ABC, giving your answer in the form $p + q\sqrt{2} + r\sqrt{3}$, where p, q, and r are integers.

Perimeter = ... cm **[3]**

11 Do not use a calculator for this question.
All working must be clearly shown.

(a) Show that

$$\frac{1-\tan^2 30^\circ}{1+\tan^2 30^\circ} = \cos 60^\circ$$

[3]

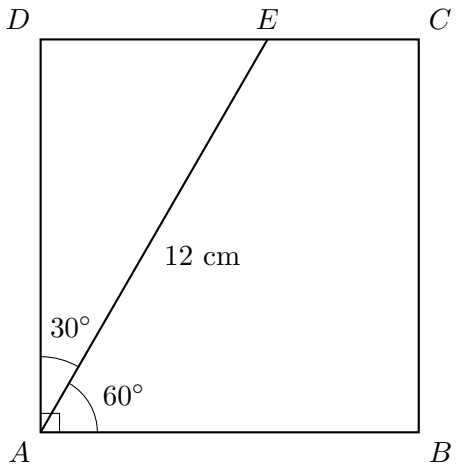
(b) Solve the equation.

$$x \sin 60^\circ = \tan 60^\circ - x \cos 60^\circ$$

Give your answer in the form $a - \sqrt{b}$, where a and b are integers.

$x =$ ..... **[4]**

12 The diagram shows a rectangle $ABCD$.
E is a point on the side CD.
$AE = 12$ cm, angle $DAE = 30°$ and angle $EAB = 60°$.



(a) Show that the exact length of AD is $6\sqrt{3}$ cm.

[2]

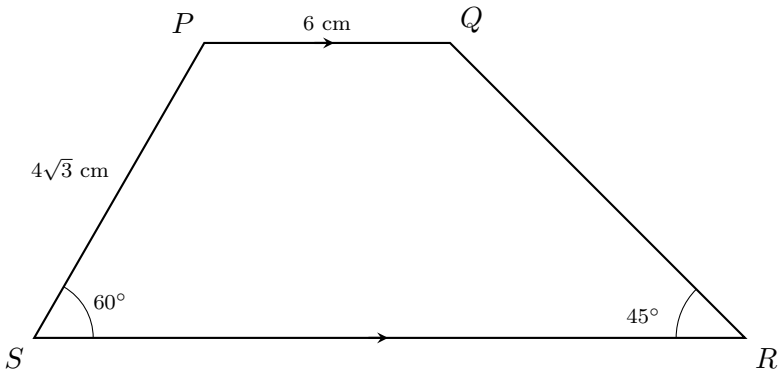
(b) The exact area of the rectangle $ABCD$ is $60\sqrt{3}$ cm^2.
Work out the exact length of CE.

..... cm **[3]**

(c) Find the exact value of $\tan(\angle EBC)$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$ where a, b, and c are integers.

..... **[3]**

13 The diagram shows a trapezium $PQRS$ with PQ parallel to SR.
$PQ = 6$ cm, $PS = 4\sqrt{3}$ cm, $\angle PSR = 60°$ and $\angle QRS = 45°$.



(a) Find the exact perpendicular height of the trapezium $PQRS$.

Height = ... cm **[2]**

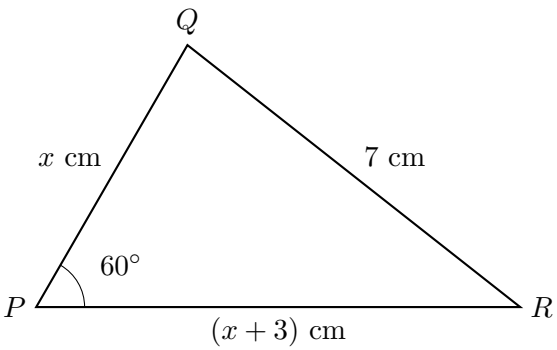
(b) Calculate the exact area of the trapezium $PQRS$.
Give your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Area = ... cm^2 **[4]**

(c) Show that the exact perimeter of the trapezium $PQRS$ is $(18 + 6\sqrt{3} + 6\sqrt{2})$ cm.

[2]

14 The diagram shows a triangle PQR.
$PQ = x$ cm, $PR = (x + 3)$ cm and $QR = 7$ cm.
Angle $QPR = 60°$.



(a) Show that $x^2 + 3x - 40 = 0$.

[4]

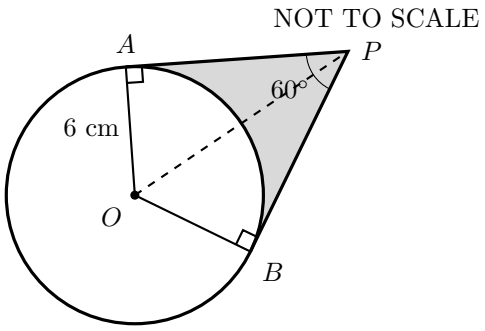
(b) Solve the equation to find the value of x, and hence find the perimeter of triangle PQR.

Perimeter = ... cm **[3]**

(c) Find the exact area of triangle PQR.

Area = ... cm^2 **[3]**

15 The diagram shows a circle, centre O, of radius 6 cm. From an external point P, tangents PA and PB are drawn to touch the circle at A and B respectively, such that angle $APB = 60°$.
The region bounded by PA, PB, and the minor arc AB is shaded.



(a) Find the exact length of the tangent PA, giving your answer in the form $k\sqrt{3}$, where k is an integer.

$PA =$ ..... cm **[3]**

(b) Calculate the exact area of the shaded region, giving your answer in the form $a\sqrt{3} - b\pi$, where a and b are integers.

Area = ... cm^2 **[4]**

(c) Calculate

(i) the exact length of the chord AB, giving your answer in the form $m\sqrt{3}$, where m is an integer,

$AB =$ ..... cm **[2]**

(ii) the exact shortest distance from point P to the chord AB.

Distance = ... cm **[2]**

1 **(a)** $\frac{\sqrt{3}}{2}$ **[B1]**

(b) $\sin 30° = \frac{1}{2}$ and $\tan 60° = \sqrt{3}$ **[M1]**

$\frac{1 - 1/2}{(\sqrt{3})^2} = \frac{1/2}{3} = \frac{1}{6}$ with no errors seen **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 2 **[B1]**

(b) $\frac{1}{2}$ **[B1]**

(c) $\frac{\sqrt{3}}{\sqrt{3}/2}$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ and $\tan 30° = \frac{1}{\sqrt{3}}$ (or $\frac{\sqrt{3}}{3}$) **[M1]**

1 **[A1]**

(b) Substitutes $\tan 45° = 1$ leading to $\cos x = \frac{1}{\sqrt{2}}$ **[M1]**

45 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $\frac{1}{2} + \frac{1}{2} = 1$ **[B1]**

(b) States at least two exact values correctly: $\tan 60° = \sqrt{3}$, $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 0° = 1$ **[B1]**

Substitutes into expression: $\sqrt{3} \times \frac{\sqrt{3}}{2} - 1 = \frac{3}{2} - 1$ **[M1]**

$\frac{1}{2}$ (or 0.5) **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** **(i)** $\sin 30° = \frac{1}{2}$ and $\cos 60° = \frac{1}{2}$ **[M1]**

1 **[A1]**

(ii) $\tan 60° = \sqrt{3}$ and $\cos 30° = \frac{\sqrt{3}}{2}$, giving $\frac{\sqrt{3}}{\frac{\sqrt{3}}{2}} = 2$ **[B1]**

$(\sin 45°)^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$ **[B1]**

$\frac{3}{2}$ (or 1.5 or $1\frac{1}{2}$) **[A1]**

(b) $\cos x = \frac{\sqrt{3}}{2}$ **[M1]**

30 **[A1]**

(c) $\tan y = 1 - 1 = 0$ **[M1]**

0 **[A1]**

[Back to Question] *[Total Marks: 9]*

6 **(a)** $\sin 45° = \frac{6}{PR}$ (or $\cos 45° = \frac{6}{PR}$) **[M1]**

$PR = 6 \div \frac{\sqrt{2}}{2} = 6\sqrt{2}$ **[A1]**

(b) $\sin 60° = \frac{6\sqrt{2}}{PS}$ **[M1]**

$PS = 6\sqrt{2} \div \frac{\sqrt{3}}{2}$ **[M1]**

$PS = \frac{12\sqrt{2}}{\sqrt{3}} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** $\sin 45° = \frac{PQ}{8\sqrt{2}}$ **[M1]**

$PQ = 8\sqrt{2} \times \frac{1}{\sqrt{2}} = 8$ **[A1]**

(b) $\cos 30° = \frac{8\sqrt{2}}{PS}$ or $PS = \frac{8\sqrt{2}}{\cos 30°}$ **[M1]**

$PS = \frac{8\sqrt{2}}{\sqrt{3}/2} = \frac{16\sqrt{2}}{\sqrt{3}}$ **[M1]**

Rationalises the denominator to get $\frac{16\sqrt{6}}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

8 In $\triangle ABD$, $AD = 8\sqrt{2}\cos 45° = 8$ **[M1]**

$BD = 8\sqrt{2}\sin 45° = 8$ **[A1]**

In $\triangle BDC$, $BC = \frac{BD}{\sin 30°} = \frac{8}{0.5} = 16$ **[M1]**

$DC = \frac{BD}{\tan 30°} = 8\sqrt{3}$ **[M1]**

Perimeter $= AB + BC + AD + DC = 8\sqrt{2} + 16 + 8 + 8\sqrt{3}$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Uses $\sin 45° = \dfrac{BD}{8}$ with $\sin 45° = \dfrac{\sqrt{2}}{2}$ (or $\dfrac{1}{\sqrt{2}}$) **[M1]**

$4\sqrt{2}$ **[A1]**

(b) Uses $\sin 30° = \dfrac{4\sqrt{2}}{BC}$ with $\sin 30° = \dfrac{1}{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) Finds $AD = 8\cos 45° = 4\sqrt{2}$ and $DC = \dfrac{4\sqrt{2}}{\tan 30°}$ (or $8\sqrt{2}\cos 30°$) **[M1]**

Evaluates $DC = 4\sqrt{2} \times \sqrt{3} = 4\sqrt{6}$ and sums $AC = AD + DC = 4\sqrt{2} + 4\sqrt{6}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** $BD = 8\sqrt{2}\sin 45° = 8\sqrt{2} \times \dfrac{\sqrt{2}}{2} = 8$ cm **[M1]**

$AD = 8\sqrt{2}\cos 45° = 8\sqrt{2} \times \dfrac{\sqrt{2}}{2} = 8$ cm **[A1]**

(b) $\tan 30° = \dfrac{8}{CD}$ giving $\dfrac{1}{\sqrt{3}} = \dfrac{8}{CD}$ **[M1]**

$8\sqrt{3}$ **[A1]**

(c) Base $AC = 8 + 8\sqrt{3}$ cm **[B1]**

Area $= \dfrac{1}{2}(8 + 8\sqrt{3}) \times 8$ **[M1]**

$32 + 32\sqrt{3}$ **[A1]**

(d) $\sin 30° = \dfrac{8}{BC}$ giving $BC = \dfrac{8}{\frac{1}{2}} = 16$ cm **[M1]**

Perimeter $= 8\sqrt{2} + 16 + (8 + 8\sqrt{3})$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

11 **(a)** Substitutes $\tan 30° = \dfrac{1}{\sqrt{3}}$ to get $\tan^2 30° = \dfrac{1}{3}$ **[B1]**

Evaluates numerator as $1 - \dfrac{1}{3} = \dfrac{2}{3}$ AND denominator as $1 + \dfrac{1}{3} = \dfrac{4}{3}$ **[M1]**

Simplifies $\dfrac{2/3}{4/3} = \dfrac{1}{2}$ which is equal to $\cos 60°$ **[A1]**

(b) Substitutes exact values: $x\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} - x\left(\frac{1}{2}\right)$ **[M1]**

Rearranges to collect x terms: $x\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \sqrt{3}$ **[M1]**

Isolates x: $x = \frac{2\sqrt{3}}{\sqrt{3}+1}$ **[M1]**

Rationalises denominator: $\frac{2\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{6-2\sqrt{3}}{2} = 3 - \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

12 (a) $\cos 30° = \frac{AD}{12}$ **[M1]**

$AD = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3}$ with no errors seen **[A1]**

(b) Uses Area $= AB \times AD = 60\sqrt{3}$ to find $AB = 10$ **[B1]**

$DE = 12\sin 30° = 6$ **[M1]**

$CE = CD - DE = 10 - 6 = 4$ **[A1]**

(c) Identifies $BC = 6\sqrt{3}$ and $CE = 4$ **[M1]**

$\tan(\angle EBC) = \frac{CE}{BC} = \frac{4}{6\sqrt{3}}$ **[M1]**

$\frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

13 (a) Uses $h = 4\sqrt{3}\sin 60°$ with $\sin 60° = \frac{\sqrt{3}}{2}$ **[M1]**

6 **[A1]**

(b) Finds left base segment: $4\sqrt{3}\cos 60° = 4\sqrt{3} \times \frac{1}{2} = 2\sqrt{3}$ **[B1]**

Finds right base segment: $\frac{6}{\tan 45°} = 6$, giving total base $SR = 2\sqrt{3} + 6 + 6 = 12 + 2\sqrt{3}$ **[B1]**

Substitutes into trapezium area formula: $\frac{1}{2}(6 + 12 + 2\sqrt{3}) \times 6$ **[M1]**

$54 + 6\sqrt{3}$ **[A1]**

(c) Finds slanted side QR: $\frac{6}{\sin 45°} = \frac{6}{\frac{\sqrt{2}}{2}} = 6\sqrt{2}$ **[M1]**

Sums all four sides: $6 + 6\sqrt{2} + (12 + 2\sqrt{3}) + 4\sqrt{3} = 18 + 6\sqrt{3} + 6\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

14 **(a)** Uses cosine rule: $7^2 = x^2 + (x+3)^2 - 2(x)(x+3)\cos 60°$ **[M1]**

Substitutes $\cos 60° = 0.5$: $49 = x^2 + x^2 + 6x + 9 - x(x+3)$ **[M1]**

Expands and simplifies: $49 = 2x^2 + 6x + 9 - x^2 - 3x$ **[M1]**

$x^2 + 3x - 40 = 0$ **[A1]**

(b) $(x+8)(x-5) = 0$ **[M1]**

$x = 5$ (rejects $x = -8$) **[A1]**

Perimeter $= 5 + (5+3) + 7 = 20$ **[A1]**

(c) Uses sine area formula: Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

15 **(a)** Angle $OAP = 90°$ (tangent to radius $= 90°$) and angle $APO = 30°$ **[B1]**

$\tan 30° = \frac{6}{PA}$ giving $\frac{1}{\sqrt{3}} = \frac{6}{PA}$ **[M1]**

$6\sqrt{3}$ **[A1]**

(b) Area of quadrilateral $OAPB = 2 \times \frac{1}{2} \times 6 \times 6\sqrt{3} = 36\sqrt{3}\ cm^2$ **[M1]**

Angle $AOB = 360° - 90° - 90° - 60° = 120°$ **[B1]**

Area of minor sector $OAB = \frac{120}{360} \times \pi \times 6^2 = 12\pi\ cm^2$ **[M1]**

$36\sqrt{3} - 12\pi$ **[A1]**

(c) **(i)** Half-chord $= 6\sin 60° = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$ cm **[M1]**

$6\sqrt{3}$ **[A1]**

(ii) $OP = \frac{6}{\sin 30°} = 12$ cm and distance from O to $AB = 6\cos 60° = 3$ cm **[M1]**

$12 - 3 = 9$ **[A1]**

[Back to Question] *[Total Marks: 11]*